DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Communication Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Communication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTBS404 - PROBABILITY THEORY AND RANDOM PROCESSES

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 A sample space is:
A. A set of all favorable outcomes
B. A set of all possible outcomes of an experiment
C. A subset of events
D. The number of ways an event can occur

2 An event that contains no outcomes is called:
A. Sample space
B. Sure event
C. Independent event
D. Null event

3 Two events A and B are mutually exclusive if:
A. $P(A \cap B) = 0$
B. $P(A | B) = P(A)$
C. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
D. $P(B | A) = 1$

4 A random variable is defined as:
A. A function mapping sample space to outcomes
B. An outcome of a random process
C. A function that assigns a real number to each outcome in the sample space
D. A deterministic value

5 Which of the following is a discrete random variable?
A. Height of students B. Number of heads in 3 coin tosses
C. Temperature D. Speed of a moving car

6 The area under the PDF curve between two points a and b gives:
A. Expectation B. Mean value
C. $P(a \le X \le b)$ D. Variance

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7 If Z=aX+b, then the mean of Z is:
A. aE[X]+b B. aE[X]
C. a+b D. E[X]+ab

8 A random vector is defined as:
A. A scalar-valued function of a random variable
B. A deterministic vector
C. A vector whose elements are random variables
D. A probability function

9 The covariance matrix of a random vector is always:
A. Symmetric and positive definite (if non-degenerate)
B. Skew-symmetric
C. Diagonal only
D. Singular

10 A binomial distribution models:
A. The number of failures before success
B. The number of successes in a fixed number of independent Bernoulli trials
C. Continuous outcomes between 0 and 1
D. The time until the next event

11 A random process is defined as:
A. A single deterministic signal over time
B. A collection of random variables indexed by time or space
C. A function of probability
D. A distribution of values

12 A wide-sense stationary (WSS) process requires:
A. Mean to be constant and autocorrelation depends only on time difference
B. Zero mean and zero variance
C. Only variance is time-invariant
D. First and third moments are constant

Q2. Solve the following.

A) Define probability with its all approaches (numerical & classical) approches. Explain disadvantages of classicial approch of probability — 6

B) Explain the probability density function (PDF) and the cumulative distribution function (CDF) along with their properties for continuous and discrete random variables. — 6

Q3. Solve the following.

A) If X is uniformly distributed in (1,2) and $Y=X^3$, then find the mean and variance of Y. — 6

B) State and prove the statement of the Schwarz Inequality. — 6

Q4. Solve Any Two of the following.

A) State and explain the Poisson distribution. Also, write the mean and variance formula for the Poisson distribution. — 6

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B) If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective. 6

C) State and explain the terms Covariance and Correlation and write the properties. 6

Q5. Solve Any Two of the following.

A) Write a note on the law of large numbers and explain the types. 6

B) State and prove the central limit theorem. 6

C) Explain in detail axioms of probability. Find the value of k 6

x	1	2	3	4
p(x)	0.1	0.4	0.1	k

Find 1) $p(x<3)$ 2) $p(2<x=4)$

Q6. Solve Any Two of the following.

A) If $\{X(t)\}$ is WSS process with autocorrelation $R(\tau)=Ae^{-\alpha|\tau|}$. Determine the second-order moment of RV $X(9)-X(4)$. 6

B) Define the power spectral density function. Write all its properties and prove any three. 6

C) Define random process. Provide the classification of random process based on nature of space and time parameter along with example for each type. 6

* * * **End** * * *

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